

JULIAN D. SANTAMARIA P

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EDUCATION

Master of Science in Engineering (Student)

Jan 2024 - Dic 2025

Universidad de Antioquia, Medellin, Colombia

Thesis: Integration of foundation models for automatic animal identification

Bachelor of Science in Electronics Engineering (GPA 4.16/5)

Feb 2018 - Dic 2023

Universidad de Antioquia, Medellin, Colombia

Thesis: Audio vs. Visual Approach to Monitor the Critically Endangered Species *Atlapetes blancae*: Developing Deep Learning Models with Limited Data

ACADEMIC EXPERIENCE

Intern

Télécom Paris, Institut Polytechnique de Paris, France.

Jan 2025 – Jun 2025

Research on domain adaptation in computer vision for biodiversity monitoring.

Supervised by Jhony H. Giraldo.

Researcher

SISTEMIC Research Group, Universidad de Antioquia, Colombia.

Jan 2024 – Present

Identification of the transformation in the Colombian forest based on acoustic analysis.

Junior Researcher

SISTEMIC Research Group, Universidad de Antioquia, Colombia.

Jan 2022 – Dec 2023

Identification of the transformation in the Colombian forest based on acoustic analysis.

Teaching Assistant

Universidad de Antioquia, Medellín, Colombia.

Jan 2024 – Present

Support for the undergraduate course “Computer Science I”.

Universidad de Antioquia, Medellín, Colombia.

Jan 2021 – Dec 2023

Academic tutoring in the course “Fundamentals of Digital Electronics”.

Visiting Research Scholar

CinfonIA Research Group, Universidad de los Andes, Colombia.

Jul 2025 – Aug 2025

Research on wildlife image segmentation.

Supervised by Pablo Arbelaez.

RESEARCH INTERESTS

Domain Generalization, Domain Adaptation, Wildlife Monitoring, Multimodal Learning, Graph Neural Networks, Computer Vision, Machine Learning.

PUBLICATIONS

- **Julian D. Santamaria**, Claudia Isaza, Jhony H. Giraldo, “*CATALOG: A Camera Trap Language-guided Contrastive Learning Model*”, **Oral Presentation** (Around 8% of all submitted papers) at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025, a Rank A conference. 
- **Julian D. Santamaria P**, Jhony H. Giraldo, Angelica Diaz-Pulido, Claudia Isaza “*Audio vs. Visual Approach to Monitor the Critically Endangered Species *Atlapetes blancae*: Developing Deep Learning Models with Limited Data*”, The 2024 IARIA Annual Congress on Frontiers in Science, Technology, Services, and Applications, 2024. 

PAPERS UNDER REVIEW

- **Julian D. Santamaria**, Claudia Isaza, Jhony H. Giraldo. “*WildIng: A Wildlife Image Invariant Representation Model for Geographical Domain Shift*”. Submitted to the International Journal of Computer Vision (IJCV).

POSTER PRESENTATIONS

- **Audio and Video Analysis using Machine Learning for Monitoring the critically endangered species *Atlapetes blancae***, International Conference on Computer Vision - ICCV 2023 - LatinX Workshop. 2023.

FELLOWSHIPS AND SCHOLARSHIPS

- **NVIDIA Academic Grant**, NVIDIA AI *Bridging the Gap between Foundation Models and Camera Trap Image Recognition*. **Role:** Recipient. **10,000 A100 GPU-Hours**. Awarded for research proposal co-written with my advisor **Jhony H. Giraldo** Jan 2025 - Jun 2025
- **Estudiante Instructor Scholarship**, Universidad de Antioquia *Funding for the Master in Engineering to support the thesis “Integración de modelos de fundación para la identificación automática de animales.”* **Role:** Recipient. **€20,450**. 2024 - 2025.
- **CV Foundation Travel Grant**, International Conference on Computer Vision (ICCV) *Scholarship to support travel and attendance at ICCV 2023*. **Role:** Recipient. **€1,000**. 2023
- **LatinX in AI Travel Scholarship**, LatinX in AI *Grant awarded to support presentation in the LatinX in AI Workshop at ICCV 2023*. **Role:** Recipient. **€400**. 2023
- **Colombian Government Undergraduate Scholarship**, Ministerio de Educación Nacional *Funding for undergraduate studies in Electronic Engineering*. **Role:** Recipient. **€17,296**. 2018 - 2023.

AWARDS AND HONORS

- **Academic Excellence Award**, Universidad de Antioquia *Awarded to the highest-performing first-level student in the Electronics Engineering program*. **Semester: 2018-1**.
- **Academic Excellence Award**, Universidad de Antioquia *Awarded to the highest-performing ninth-level student in the Electronics Engineering program*. **Semester: 2023-1**.

PROFESSIONAL SERVICE

Journal Reviewer: IEEE Transactions on Biomedical Engineering and EURASIP Journal on Image and Video Processing.

Workshop Reviewer: NeurIPS 2025 Workshop NPGML.

COMPUTER SKILLS

Programming Languages

Python, Matlab, C/C++.

Relevant Libraries

PyTorch, PyTorch Geometric, OpenCV, Transformers (HuggingFace).

Tools & Frameworks

Weights & Biases (wandb), Git, L^AT_EX, Inkscape.

LANGUAGES

Spanish: Native Language.

English: B2. IELTS: 5.5/9 (2017), Professional Working Proficiency.